TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 1 Estimation of Characteristic Impedance (Z_o) and Propagation Constant (γ) of a Transmission Line

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AIM:

Investigating the significance of Characteristic Impedance (Zo), and Propagation

Constant (γ) of a Transmission Line with Resistance =10 Ω/Km, Capacitance = 8 μF/Km, Inductance =3.5 mH/Km, Conductance =4 μmho/Km, Frequency= 2 Khz

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

THEORY / FORMULAE

Characteristic impedance:

$$Z_o = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$$

Propagation constant:

$$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$$

where α is the attenuation constant in neper/km, β is the phase constant in radians/km, and $\omega = 2\pi f$.

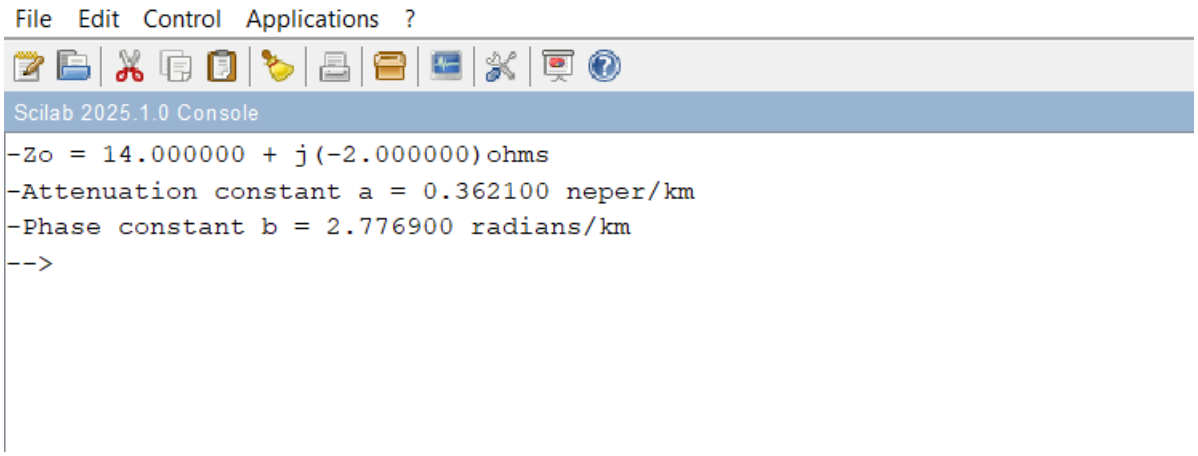
PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT :



RESULTS

Test Case 1: The attached SCILAB output gives $Z_o = 14 - j2\ \Omega$, $\alpha = 0.3621$ neper/km and $\beta = 2.7769$ radians/km.

Test Case 2: The effect of resistance and inductance on Z_o, α and β was evaluated for the ranges specified in the manual.

Test Case 3: The effect of frequency and capacitance on Z_o, α and β was evaluated for the ranges specified in the manual.